

Frontier League Single-Variable Regression Report

Analysis mode: both
Dependent variable: WinPct -> Pitching_W-L%
Independent variable: OPS -> Batting OPS
Rows used: 80
PDF location: C:\Users\Matthew\OneDrive\Desktop\frontier league regression analysis\frontier_single_variable_regression_report.pdf

Standard Regression Summary

Formula: Q("Pitching_W-L%") ~ Q("Batting_OPS")

Coefficient Table						
	coef	std_err	t_stat	p_value	ci_lower	ci_upper
term						
Intercept	-0.4303	0.1281	-3.3578	0.0012	-0.6854	-0.1752
OPS	1.2266	0.1684	7.2844	0.0000	0.8914	1.5618

Overall Model Fit	
	value
R-squared	0.4049
Adjusted R-squared	0.3972
F-statistic	53.0619
F-statistic p-value	0.0000
Observations	80.0000

Heteroskedasticity Tests					
Test	LM Statistic	LM p-value	F Statistic	F p-value	
Breusch-Pagan	2.8917	0.089	2.9251	0.0912	
White	8.0351	0.018	4.2986	0.0170	

Interpretation

- Higher OPS is associated with higher WinPct. The estimated coefficient is 1.2266 and the relationship is statistically significant (p = 0.0000). A one-unit increase in OPS changes WinPct by about 1.2266.
- The model R-squared is 0.405, so the selected regressor explains about 40.5% of the variation in WinPct. This is modest explanatory power.
- HC1 robust standard errors were used for inference because the heteroskedasticity tests suggested non-constant error variance.

Jarque-Bera

JB statistic = 3.8020, p-value = 0.1494, skewness = -0.5322, kurtosis = 3.0868

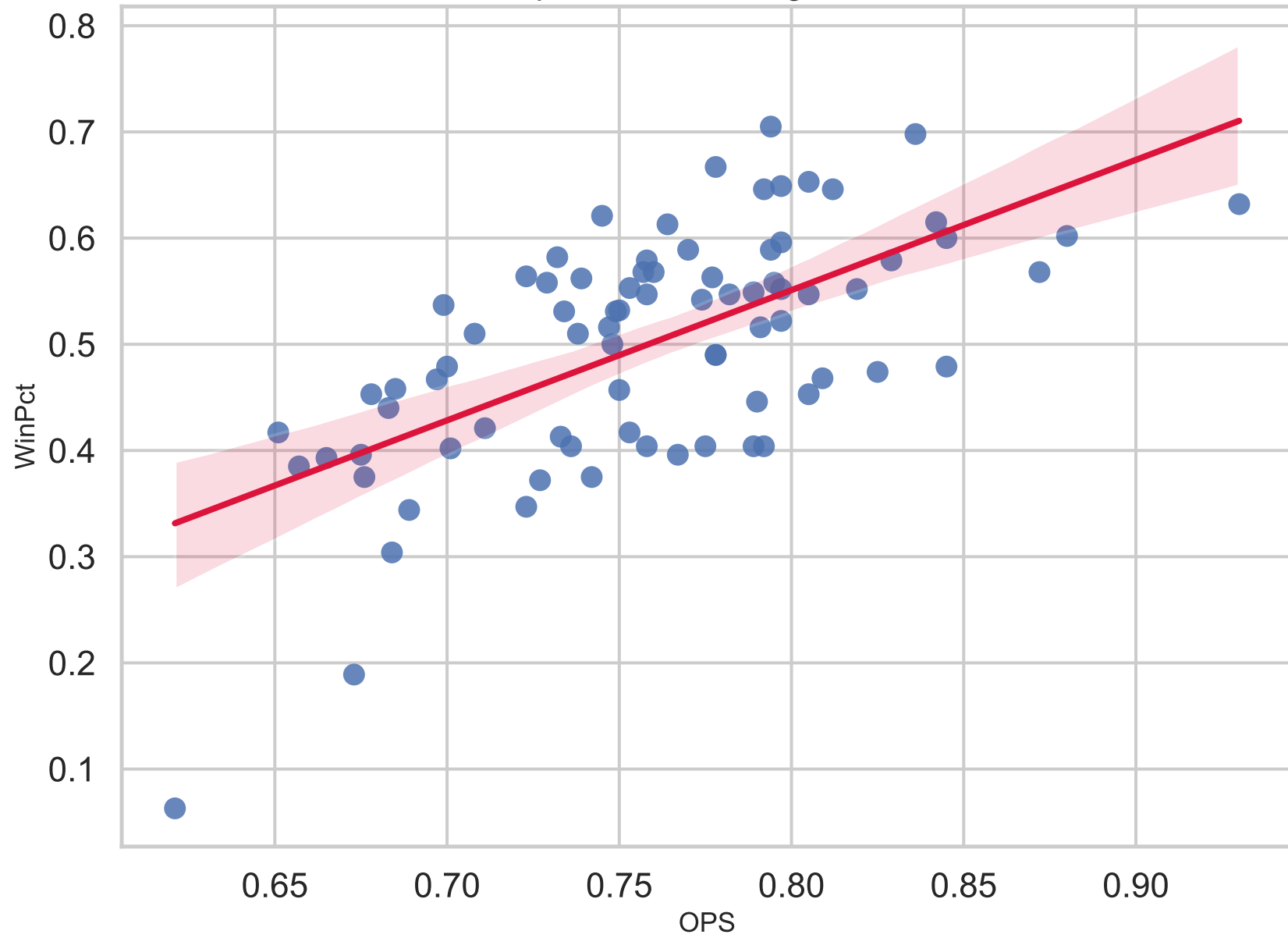
Reverse Regression Summary

Formula: Q("Batting_OPS") ~ Q("Pitching_W-L%")

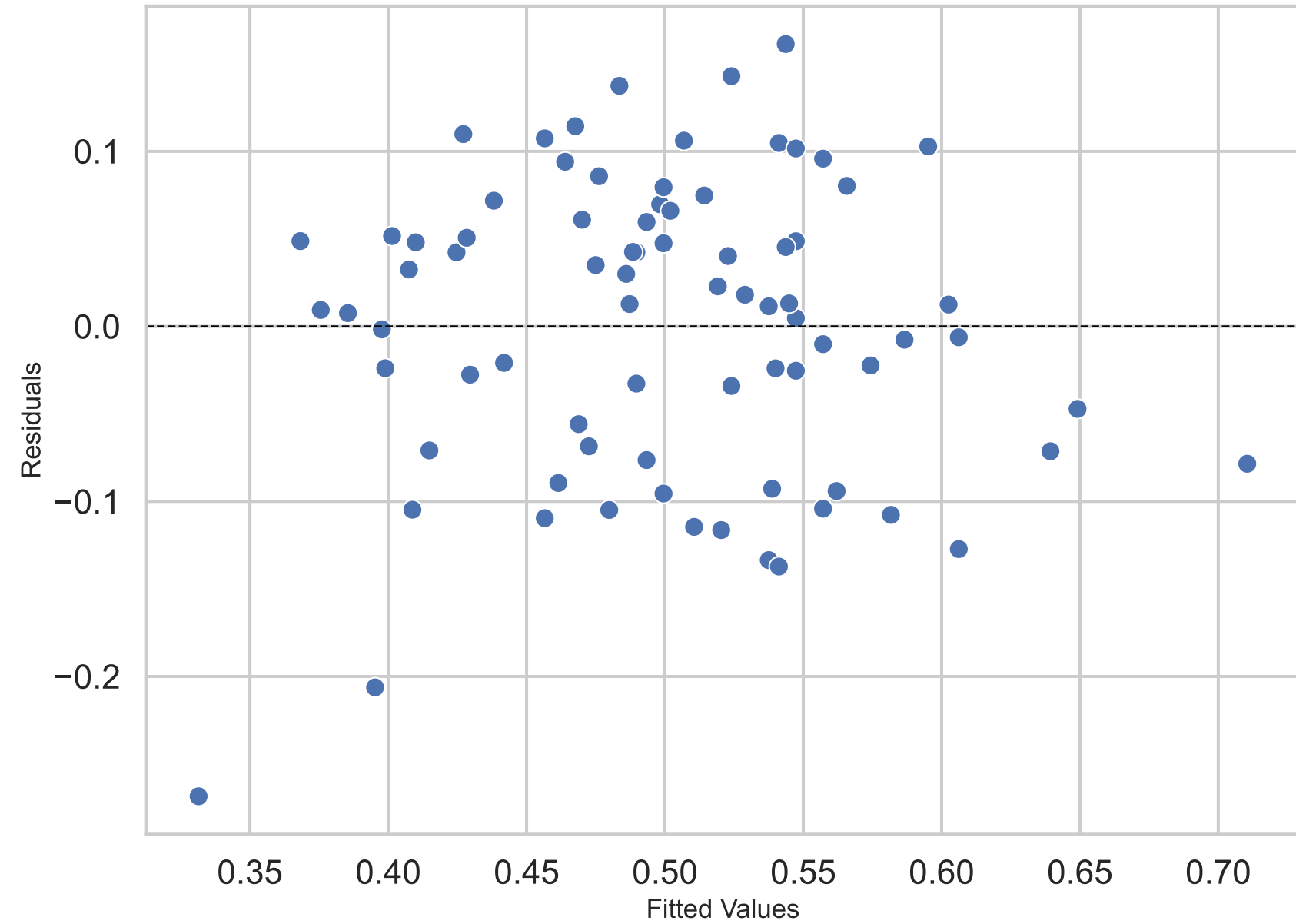
Reverse Coefficient Table						
	coef	std_err	t_stat	p_value	ci_lower	ci_upper
term						
Intercept	0.5937	0.0232	25.5757	0.0	0.5474	0.6399
WinPct	0.3301	0.0453	7.2844	0.0	0.2399	0.4203

Reverse Model Fit		value
R-squared		0.4049
Adjusted R-squared		0.3972
F-statistic		53.0619
F-statistic p-value		0.0000
Observations		80.0000

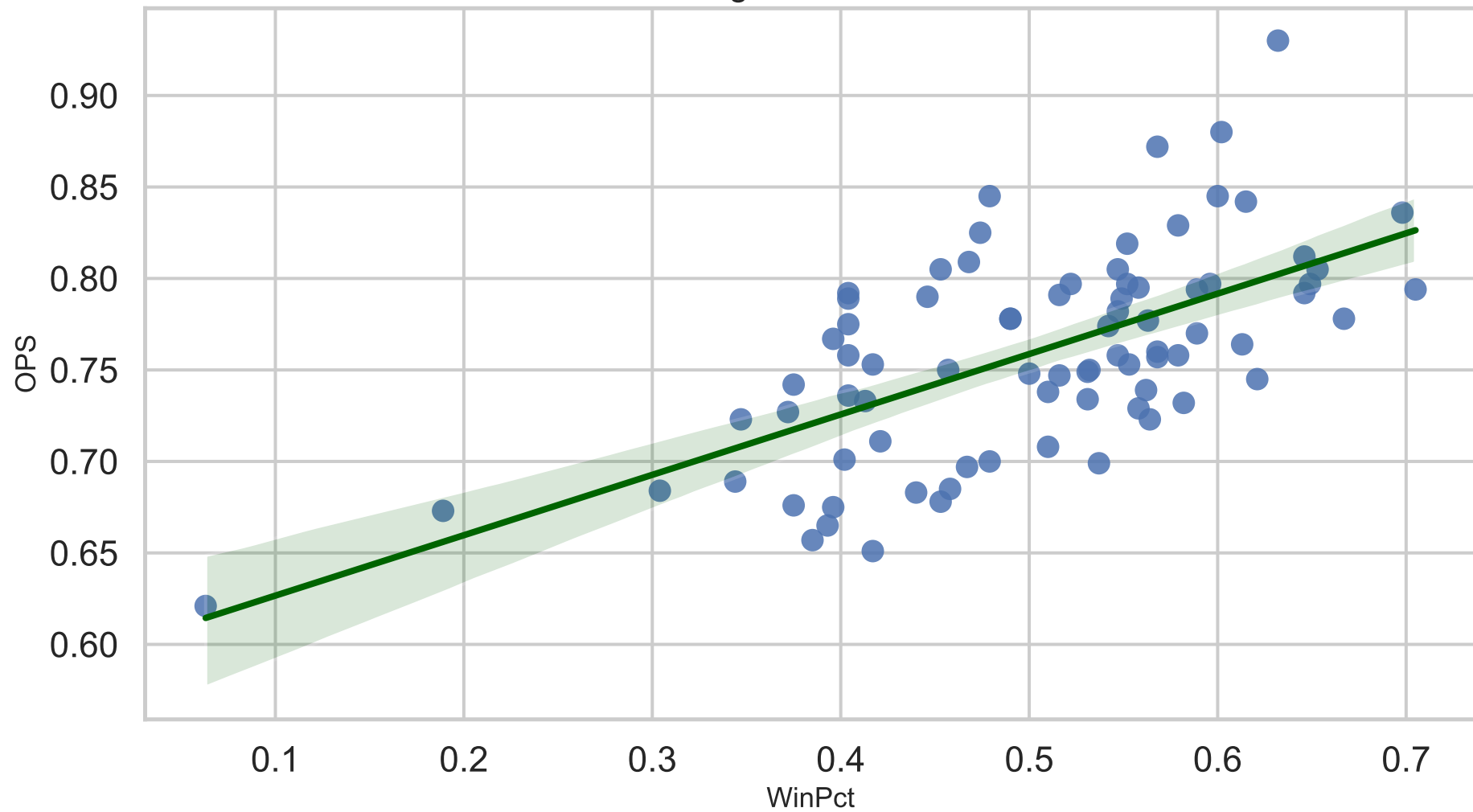
Scatterplot with OLS Regression Line



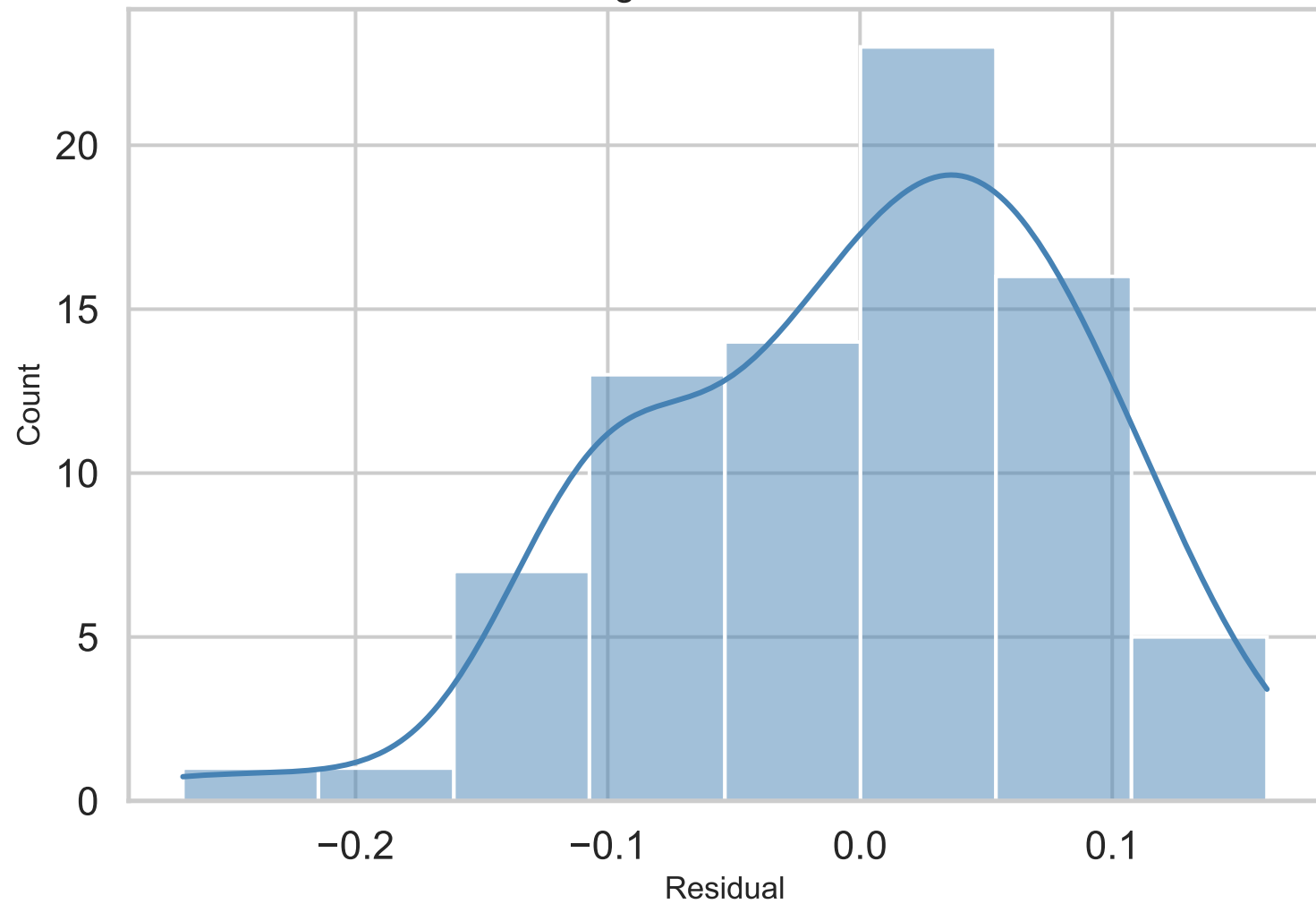
Residual Plot



OPS Regressed on WinPct



Histogram of Residuals



QQ Plot of Residuals

